Ethan Yap

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SUMMARY

IT student seeking a 44-week software engineering internship (15 Mar 2027 – 14 Jan 2028). Builds and deploys full-stack web apps with Node.js, Next.js, Spring Boot and Python on PostgreSQL. Main projects ship with automated tests run in GitHub Actions CI.

EDUCATION

Singapore Polytechnic · Diploma in IT (Software and Applications), Minor in Business — Apr 2025 – Apr 2028 (expected)
Cumulative GPA 3.43 / 4.00 · Key modules: Back-End Web Development, Java Programming, Continuous Integration and Continuous Delivery, Design for User Interaction

Maris Stella High School · GCE O-Level, L1R5 net 10 · Direct School Admission (Robotics) — 2021 – 2024

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, Java, SQL
Backend	Node.js, Express, Spring Boot, FastAPI, REST APIs, JWT auth, WebSockets
Frontend	React, Next.js, Tailwind CSS
Data & AI	PostgreSQL, pgvector, MySQL, pandas, embeddings, RAG
DevOps & Testing	Docker Compose, GitHub Actions (CI/CD), Vercel, Git, pytest, node:test
Spoken	English, Mandarin

PROJECTS

Incident Monitoring Dashboard · *Docker Compose, Node.js, PostgreSQL, WebSockets, Next.js* — Code · Demo

- Designed a 5-container Docker Compose stack with health checks, verified end to end by a smoke test in GitHub Actions.
- Built an alerting engine that opens after 2 bad readings and resolves after 3 good ones, filtering noisy spikes; 15 unit tests.

AI Code Snippet & Documentation Hub · *Next.js, TypeScript, PostgreSQL, pgvector, Gemini API* — Code · Live · Demo

- Kept saves instant with a background worker on a Postgres SKIP LOCKED job queue that writes Zod-validated LLM docs.
- Built semantic search over 768-dimension embeddings using a pgvector HNSW index, re-ranked by full-text score; 73 tests.

Slotted: Facility Booking System (in progress) · *Spring Boot, Java 21, PostgreSQL, React* — Code

- Prevented double bookings with a Postgres exclusion constraint, proven by concurrency tests among 184 API tests in CI.
- Documented the design before building: PRD, ERD, OpenAPI contract and architecture decision records.

Academic Resource RAG Search Engine · *Python, FastAPI, ChromaDB, pytest* — Code · Demo

- Built PDF ingestion that chunks by sentence, page and section, so every answer cites its exact page.
- Refused off-topic questions with a relevance cut-off and rejected uncited LLM answers, covered by pytest tests in CI.

Pokémon Gym & Battle System · *Node.js, Express, PostgreSQL, JWT* — Code · Live · Demo

- Built a 49-endpoint REST API with JWT auth and ownership checks, then migrated it from MySQL to PostgreSQL.
- Wrote a battle engine and a transactional gacha shop, covered by 20 API tests in CI against a real PostgreSQL.

More projects (Steady calorie tracker, Tech Trend Sentiment Tracker, MyLearningAI): ethanyap123.github.io/portfolio

WORK EXPERIENCE

Two Men Bagel House · Line Chef (part-time) — Feb 2025 – Apr 2026

- Coordinated orders between kitchen and front of house at peak service; also ran the register and barista station.

Bee Cheng Hiang · Barbecue Cook (part-time) — Nov 2024 – Feb 2025

- Ran a cooking station solo through the festive peak, preparing ahead of demand to shorten queues.

LEADERSHIP, ACTIVITIES & AWARDS

SP School of Computing CCAs · Member, Apple Developers Society (ADS) and SEED (Software Development SIG); helped run multiple ADS and SEED workshops for members — 2025 – present

SP student volunteer · SP Open House; event helper at School of Computing industry talks for Year 1 students — 2026

Marist Robotics & Programming Club · Mentored juniors; Perse Coding Team Challenge 2022–23 (**Merit Award**) — 2021 – 2024

Marist Ambassador · Emcee at school events including prize-giving; Open House volunteer — 2021 – 2024

Community service · VIA project in collaboration with Magorium (startup turning plastic waste into road material); ran registration for Mummy Yummy (charity delivering free vegetarian meals to needy households) events of 100+; volunteered at Masjid Kassim — 2022 – 2025

Edusave Character Award · **Class Chairman** (Sec 4) · **Peer Supporter** (Sec 2) — 2022 – 2024